Counting Particles with the Mole Slides · run-sheet

Period 1 · 43.75 min

1. Counting particles with the mole Lesson 5.1 · Reading §5.1 · 0.5 min

0. **enter** (0.5 min): Two periods. Read the driving question aloud; students copy it. Show the four clicker buttons once, so a zoom or a blank screen is on purpose.

Safety. Nothing dangerous in this lesson. A real bag of nickels and a balance may be used live as the hook, and are recommended; nothing in the deck depends on it. **Data and sources.** U.S. Mint coin specifications: nickel 5.000 g, cent made after 1982 2.500 g (zinc coated with copper); the cent made before 1982, 3.11 g (95% copper), from secondary sources. The Mint struck its last circulating cent on November 12, 2025, so the deck says "cents made after 1982", never "today's cents". BIPM: the 2019 definition of the mole (exactly $6.02214076 \times 10^{23}$, voted November 16, 2018, in force May 20, 2019) and the 1971 resolution (12 g of carbon-12). NIST: the silicon spheres of the Avogadro project, lattice parameter 543.102 pm. NobelPrize.org: Jean Perrin, Physics 1926. NASA JPL: the universe 13.8 billion years old. USGS: Earth 4.54 billion years (the deck says about 4.5 billion). OpenStax Chemistry 2e, section 3.1. CIAAW masses: Al 26.982, Cu 63.546, Si 28.085 (g for one mole). Shannon 1976 ionic radii (Na^+ 102 pm, Cl^- 181 pm); Cordero 2008 covalent radii; NIST CCCBDB geometries (H_2O 104.5°, CO_2 180°, NH_3 106.7°); NaCl cell 564 pm (the value Unit 4 used). **Clicker.** Logitech R800: Next = PageDown, Back = PageUp (while zoomed: exit the zoom), Start/stop = F5 then Escape, Blank = period. Blank zooms one part of the slide per press, animation first; the press after the last part exits; two fast presses on an unzoomed slide blank the screen. Mouse: click an option to mark it right or wrong; click the slide number or the section label for the slide list. Laptop: N notes, O overview, T theme, Ctrl+P run-sheet.

2. The coin jar Hook · Reading §5.1.1 · 2.5 min

0. **enter** (0.5 min): Ask: who has a jar of coins at home? How would you find out how many are in it?
1. **predict** (0.5 min): Prediction: each table writes one guess and one way to count. Collect the guesses on the board. Do not judge them.
2. **count one by one** (0.5 min): The counter climbs one coin a second, then "lost count". Ask: how long would a jar of 400 take? Listen for "too long".
3. **the scale** (0.25 min): The scale reads the coins: 2,000.0 g. Every nickel has a mass of 5.000 g.
4. **divide** (0.25 min): 2,000.0 g ÷ 5.000 g = 400 nickels. Compare with the guesses. Misconception to listen for: "multiply the two masses".
5. **the thread** (0.5 min): The thread line. Students write it at the top of the page: too many to count? Weigh them.

Narrative thread for 5.1: too many to count? weigh them, then think in moles. An everyday jar, no chemistry yet. The bank's bag comes next (slide 3), and the bank's scale returns on slide 6.

3. A bag of nickels at the bank Launch · Reading §5.1 opener · 3 min

0. **enter** (0.5 min): The bank never opens the bag. Ask: how can it know there are 250?
1. **a spoonful of water** (0.5 min): About 18 g of water, a little more than a spoonful. Ask: could anyone count its molecules?
2. **atoms** (0.5 min): Atoms are far too small to see, and far too many to count.
3. **driving question** (0.5 min): Read the driving question. Students copy it at the top of the notebook page.
4. **write 60 s** (1 min): Sixty seconds with the starter. Collect two answers; do not correct yet. Listen for "a microscope" and "a very big number".

The reading's opener: a bank teller counts a sealed bag of nickels by weighing it. A few grams of any substance hold more atoms than anyone could count one by one.

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4. Predict now Predict · Reading §5.1 Before you read · 2 min

0. **enter** (0.75 min): Count hands for A, B, C and write the counts on the board. Do not react. A mouse click on an option only highlights it; there is no grading on this slide.
1. **question 2** (0.5 min): Count hands. Many choose "molecules" for salt (the particle is a formula unit).
2. **question 3** (0.75 min): Count hands. "1 mol" is the common idea: one water, one hydrogen. Keep the counts for slide 26.

No answers on this slide. Students keep their letters; slide 26 returns all three answers.

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5. Targets and words Targets · Reading §5.1 Vocabulary · 2.5 min

0. **enter** (1 min): Read the four targets. Students copy 5.1.1 to 5.1.4; they rate themselves on slide 27.
1. **words** (1 min): Choral reading: English term, then the Spanish cognate. Students add the words to the glossary with the icon. Stress: Avogadro's number has no unit; the Avogadro constant is per mole.
2. **color code** (0.5 min): Color code: grams amber, moles teal, particles violet; cation red with +, anion blue with −; atoms in CPK colors.

Unit 5 color code (plan section 3), the same in every Unit 5 deck: mass in grams amber, amount in moles teal, number of particles violet; a conversion factor is a gray bridge; ions keep Unit 4's red "+" and blue "−"; atoms keep CPK colors with the symbol on every sphere. Shapes carry a sign or a symbol, so color-blind students can read every drawing.

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6. Counting by weighing Counting by weighing · Reading §5.1.1 · 4 min

0. **enter** (0.25 min): A bank bag of nickels on the bank's scale. Ask: what would the scale show?
1. **zero** (0.5 min): The empty bag on the scale, then zero: the reading is 0.0 g. Say "zero", the word "tare" once.
2. **predict** (0.5 min): Prediction: 1,250.0 g of nickels. How many? Collect answers on whiteboards. Listen for "multiply".
3. **divide** (0.25 min): 1,250.0 g × 1 nickel / 5.000 g: the grams cancel (struck in red).
4. **count** (0.25 min): 250 nickels. The check multiplies back: 250 × 5.000 g = 1,250.0 g.
5. **rule** (1 min): Students copy the rule card (reading §5.1.1 Key idea).
6. **check** (1 min): Hands up for A, B or C. The animation shows the roll: 200.0 g on the scale. The answer is not drawn.
7. **answer** (0.25 min): A is right: 200.0 g ÷ 5.000 g = 40. B (1,000): that multiplies the two masses; divide by the mass of one nickel. C (200): 200 is the mass in grams, not the count; divide by 5.000 g.

Zeroing (taring) in words: the scale reads 0.0 g with the empty bag on it, so it then shows the mass of the coins alone. The grams cancel, leaving a pure number of nickels.

7. Counting a mixed jar Counting by weighing · Reading §5.1.1 · 4 min

0. **enter** (0.25 min): Two kinds of cents: made before 1982 (95% copper, 3.11 g) and made after 1982 (zinc coated with copper, 2.500 g). The jar of pennies has a mass of 1,017.1 g.
1. **predict** (0.25 min): Prediction: thumbs up or down for dividing by 2.500 g. Most say up.
2. **wrong divisors** (0.25 min): ÷ 2.500 g gives about 407: too many, because some cents are heavier. ÷ 3.11 g gives about 327: too few.
3. **sample** (0.25 min): A sample of 100 cents from this jar has a mass of 267.69 g.
4. **average** (0.25 min): 267.69 g ÷ 100 = 2.6769 g per cent. Misconception: "average 2.500 and 3.11" (works only with equal numbers of each).
5. **count** (0.25 min): 1,017.1 g ÷ 2.6769 g = about 380 cents. Check: 2.6769 lies between 2.500 and 3.11.
6. **new word** (0.25 min): Read the word card; students repeat "average atomic mass, masa atómica promedio".
7. **isotopes** (0.25 min): The coins become atoms: one element, a mix of isotopes. The periodic table gives the average mass: copper 63.546. No isotope masses are needed.
8. **rule** (1 min): Students copy the rule card (reading §5.1.1 Key idea).
9. **check** (0.75 min): Hands up for A, B or C. The animation returns to the sample of 100 cents.
10. **answer** (0.25 min): A is right: weigh a sample of this jar and divide its mass by its count. B (the average of 2.500 g and 3.11 g): that works only if the jar holds equal numbers of each kind. C (2.500 g): dividing by the lighter cent counts too many coins.

Lesson 2.3 link: the average atomic mass on the periodic table is the average over an element's isotopes, just as 2.6769 g is the average cent of this jar. "About 407" (1,017.1 g ÷ 2.500 g) and "about 327" (1,017.1 g ÷ 3.11 g) are computed from the reading's data; the reading says "too many" and "too few". The two atom sizes in the isotope picture stand for two masses; real isotopes of one element have the same size.

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8. The mole The mole · Reading §5.1.2 · 3.5 min

0. **enter** (0.25 min): A counting unit: the baker counts eggs by the dozen. Ask for other counting units (a pair, a ream).
1. **a ream** (0.25 min): A ream of paper is 500 sheets. Paper is thin, so the unit is bigger.
2. **predict** (0.25 min): Prediction: how big must the counting unit be for atoms? Collect guesses: a thousand? a million?
3. **the mole** (0.25 min): One mole is 6.022×10^{23} particles. Say it: six point oh two two times ten to the twenty-third. The symbol is mol.
4. **exactly** (0.25 min): Since May 20, 2019 the number is exact: $6.02214076 \times 10^{23}$. In calculations we round it.
5. **the constant** (0.25 min): Avogadro's number has no unit; the Avogadro constant, N_A, is the same number per mole.
6. **rule** (1 min): Students copy the rule card. Misconception: "a mole is a mass".
7. **check** (0.75 min): Hands up for A, B or C. The figure of the three counting units stays on screen.
8. **answer** (0.25 min): A is right. B (a mass in grams): a mole is a count; one mole of different substances has different masses. C (a volume): one mole of different substances fills different volumes.

"Amount of substance" is the SI quantity; the mole is its unit. The rounded value 6.022×10^{23} is the exact number written with fewer digits, not a measurement. N_A is read "N sub A".

9. How many is a mole? The mole · Reading §5.1.2 · 4.25 min

0. **enter** (0.25 min): One water molecule, drawn in CPK colors: O red, H white.
1. **a dozen** (0.25 min): A dozen molecules in a carton. Twelve is easy to count.
2. **a thousand** (0.25 min): A thousand: the front face shows 100 cells; the cube is 10 deep. The counter says 1,000 = 10^3.
3. **predict** (0.25 min): Prediction: how many zoom-outs by 1,000 to reach 10^{23}? Collect numbers. Listen for "23".
4. **zoom out** (0.75 min): Each 1.2 s the whole cube shrinks into one cell of a new face: 10^6, 10^9, … 10^{21}. Read the words aloud with the class.
5. **one mole** (0.25 min): Seven steps reach 10^{21}; × 602 more gives 6.022×10^{23}: one mole, the flask of slide 8.
6. **the spoon** (0.25 min): One mole of water is only about 18 g, less than a large spoonful.
7. **rule** (1 min): Students copy the rule card. Misconception: "one mole always has a mass of 1 gram".
8. **check** (0.75 min): Hands up for A, B or C. The spoon of water stays on screen.
9. **answer** (0.25 min): A is right. B (1 g): a mole is a count, not one gram; a mole of water is about 18 g. C (6.022×10^{23} g): that is the number of molecules, not the mass in grams.

The prediction's answer: seven zoom-outs of × 1,000 reach 10^{21} (1,000 × 1,000 × … seven times), then a last factor of 602: $10^{21} \times 602.2 = 6.022 \times 10^{23}$. The step words (a million, a billion, …) are names of powers of ten. Under reduced motion the cascade shows its end state with the ladder complete.

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10. Name the particle The mole · Reading §5.1.2 · 3.75 min

0. **enter** (0.25 min): One mole of aluminum atoms on the left pan: 26.982 g. The beam tips toward the loaded pan.
1. **copper** (0.25 min): One mole of copper atoms: 63.546 g. The beam tips toward copper.
2. **predict** (0.25 min): Prediction: which pan holds more atoms? Hands up for aluminum, copper, the same.
3. **same number** (0.25 min): Both hold 6.022×10^{23} atoms. A copper atom has more mass. Misconception: "a mole is a mass".
4. **predict** (0.25 min): Prediction: "one mole of oxygen" could mean either. Which?
5. **oxygen** (0.25 min): 1 mol of O_2 molecules holds 2 mol of O atoms: twice the atoms of 1 mol of O atoms.
6. **four kinds** (0.25 min): Atoms (a Cu atom), molecules (H_2O), formula units (one Na^+ with one Cl^-), ions. Read the caption: atom colors show the element, not the charge.
7. **rule** (1 min): Students copy the rule card (reading §5.1.2: a mole must always name its particles).
8. **check** (0.75 min): Hands up for A, B or C. The balance returns: one mole of Al, one mole of Cu.
9. **answer** (0.25 min): A is right. B (the same mass): each element's atoms have their own mass, so the moles differ in mass. C (different numbers; copper atoms are larger): a mole always holds 6.022×10^{23} particles, whatever their size.

Masses of one mole from the blueprint's misconception table (CIAAW): aluminum 26.982 g, copper 63.546 g. Atom sizes from covalent radii (Cordero 2008); ion sizes from Shannon 1976. The caption under the picture says that atom colors (CPK) show the element; ions keep Unit 4's red "+" and blue "−".

11. The size of a mole The mole · Reading §5.1.2 · 4 min

0. **enter** (0.25 min): The counter ticks one particle each second beside the clock. Ask: how long to count a mole this way?
1. **one year** (0.25 min): One year is about 3.156×10^{7} seconds. Each mark of the line is ten times the one before.
2. **Earth** (0.25 min): Earth: about 4.5 billion years (USGS 4.54 billion).
3. **the universe** (0.25 min): The universe: about 13.8 billion years (NASA JPL).
4. **predict** (0.25 min): Prediction: point at the line. Where does 6.022×10^{23} seconds land?
5. **the mole** (0.25 min): Far to the right: 6.022×10^{23} s is about 1.9×10^{16} years.
6. **how many times** (0.25 min): $1.9 \times 10^{16} \div 1.38 \times 10^{10}$ is about 1.4 million times the age of the universe.
7. **a spoonful** (0.25 min): Yet one mole of water is about 18 g: less than a large spoonful.
8. **rule** (1 min): Students copy the three lines of the rule (reading §5.1.2 Key idea).
9. **check** (0.75 min): Hands up for A, B or C. The line shows the universe bracket and, faint, the Earth bracket.
10. **answer** (0.25 min): A is right: $1.9 \times 10^{16} \div 4.5 \times 10^{9}$ is about 4×10^{6}. B (4,000 times): divide the powers of ten carefully, $10^{16} \div 10^{9} = 10^{7}$. C (1.4 million times): that is for the universe; Earth is younger, so more times.

The reading's worked example: 6.022×10^{23} s × 1 year / 3.156×10^{7} s = 1.908×10^{16} years; $1.908 \times 10^{16} \div 1.38 \times 10^{10} = 1.38 \times 10^{6}$, about 1.4 million. Each mark on the line is 10 times the one before. "Avogadro's number is so large because atoms are so small."

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12. Who found the number The mole · Reading §5.1.2 · 2.75 min

0. **enter** (0.25 min): 1811, Amedeo Avogadro: equal volumes of gases at the same temperature and pressure hold equal numbers of particles. He never knew how large that number was. Misconception: "Avogadro measured the number".
1. **1865** (0.25 min): 1865, Josef Loschmidt: the first estimate of the size of molecules, and so of the number.
2. **1890s** (0.25 min): 1890s, Wilhelm Ostwald: the word Mol as a unit.
3. **1909** (0.25 min): Around 1909, Jean Perrin measured the number by several independent methods and proposed naming it after Avogadro; Nobel Prize in Physics, 1926.
4. **1971** (0.25 min): 1971: the mole became an SI base unit, defined by 12 g of carbon-12.
5. **predict** (0.25 min): Prediction: is the number still measured today? Hands up for "measured" and for "fixed".
6. **2018 and 2019** (0.25 min): The vote of November 16, 2018; in force May 20, 2019. The number is now exact and is no longer measured.
7. **rule** (1 min): Students copy the rule card. No check on this slide.

The reading's section "How the number was found". The 1971 definition (the atoms in exactly 12 g of carbon-12) is "a definition that older books still print". The silicon spheres of the Avogadro project come next, on slide 13.

13. Counting the atoms in a silicon sphere The mole · Reading §5.1.2 · 3.5 min

0. **enter** (0.25 min): The international Avogadro project polished nearly perfect spheres of silicon. Ask: what could you measure about this ball?
1. **predict** (0.25 min): Prediction: how could anyone count its atoms? Collect ideas. Misconception: "they weighed one atom".
2. **one crystal** (0.25 min): The sphere is a single crystal: the atoms repeat in identical cubes. The model turns slowly.
3. **one cube** (0.25 min): X-rays measure the edge of one cube: 543.102 pm. Each cube holds the equivalent of 8 atoms (8 corners × 1/8, 6 faces × 1/2, 4 inside).
4. **one atom** (0.25 min): The density gives the mass of one cube; divide by 8 for the mass of one atom.
5. **the count** (0.25 min): The mass of one mole of silicon ÷ the mass of one atom = the number of atoms in one mole: 6.022×10^{23}.
6. **rule** (1 min): Students copy the rule card. The project counted atoms; it did not weigh one atom directly.
7. **check** (0.75 min): Hands up for A, B or C. The lit cube with its edge returns to the screen.
8. **answer** (0.25 min): A is right. B (define the kilogram by its size): the spheres counted atoms; since 2019 the kilogram is defined by the Planck constant. C (compare silicon and carbon atoms): the project counted atoms; it did not compare their sizes.

Honors (notes only, never assessed). The real spheres were enriched to more than 99.99% silicon-28: natural silicon is a mix of silicon-28, -29 and -30, and every uncertainty in those abundances becomes an uncertainty in the count; the other isotopes were measured with a mass spectrometer. **The model's arithmetic** with natural silicon: one cube has an edge of 543.102 pm, so a volume of $(5.43102 \times 10^{-8}\ \text{cm})^3 = 1.6019 \times 10^{-22}\ \text{cm}^3$; $\times 2.329\ \text{g/cm}^3 = 3.7308 \times 10^{-22}$ g per cube; ÷ 8 atoms = 4.6635×10^{-23} g per atom; 28.085 g ÷ 4.6635×10^{-23} g = 6.022×10^{23} atoms per mole. The real spheres used their own measured values. "Molar mass" is Lesson 5.2's word; here say "the mass of one mole of silicon".

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14. One picture: three ways to count Summary · Reading §5.1.1 and §5.1.2 · 3.5 min

0. **enter** (0.25 min): The empty table. Ask: what were the three things we counted today?
1. **nickels** (0.25 min): Nickels: every nickel has the same mass.
2. **cents** (0.25 min): Cents: two kinds, two masses.
3. **atoms** (0.25 min): Atoms: far too small and far too many.
4. **the problem** (0.25 min): The problem in each column.
5. **the trick** (0.25 min): The trick: divide by one mass, by the average mass, or count in moles of 6.022×10^{23}.
6. **name it** (0.25 min): Name it: a mole always names its particles.
7. **the thread** (0.25 min): The thread line, said together: too many to count? Weigh them. Then think in moles.
8. **write 90 s** (1.5 min): Ninety seconds of writing with the frames and the word bank. End of period 1.

One picture closes period 1. Students copy the three columns as one diagram. Period 1 ends here: collect three notebooks as a quick check before period 2.

Period 2 · 46 min

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15. Warm-up Warm-up, period 2 · Reading §5.1.1 and §5.1.2 · 2.5 min

0. **enter** (0.25 min): Students read the questions while you take attendance.
1. **timer 90 s** (1.5 min): Ninety seconds. Students answer in the notebook.
2. **answer 1** (0.25 min): 1,000.0 g ÷ 5.000 g = 200 nickels. Misconception: "multiply".
3. **answer 2** (0.25 min): 6.022×10^{23}, exactly $6.02214076 \times 10^{23}$ since 2019.
4. **answer 3** (0.25 min): Atoms, molecules, formula units, ions. A mole always names its particle.

Period 2 starts here. The three questions retrieve 5.1.1 and 5.1.2. 1,000.0 g ÷ 5.000 g = 200 is arithmetic on the reading's data.

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16. Moles to particles Moles and particles · Reading §5.1.3 · 3.25 min

0. **enter** (0.25 min): The map: moles (teal) on the left, particles (violet) on the right, the gray bridge 6.022×10^{23} per mole.
1. **predict** (0.25 min): Prediction: 2.00 mol of copper atoms. More or fewer than 6.022×10^{23}? Listen for "fewer" (dividing).
2. **setup** (0.25 min): Write the factor with its units: 6.022×10^{23} atoms over 1 mol.
3. **cancel** (0.25 min): mol cancels (struck in red); atoms remain (circled).
4. **result** (0.25 min): 1.20×10^{24} Cu atoms. Check: more than one mole, so more than 6.022×10^{23}.
5. **rule** (1 min): Students copy the rule card. Misconception: "divide to get particles".
6. **check** (0.75 min): Hands up for A, B or C. The copper example stays on the map.
7. **answer** (0.25 min): A is right. B (divide): dividing gives a tiny number, smaller than one particle; check the units. C (the mass of one particle): that would give a mass; to count particles, use the Avogadro constant.

Reading §5.1.3: the Avogadro constant is a conversion factor; multiply by 6.022×10^{23} particles/mol so that mol cancels. Copper and other metals are counted as atoms.

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17. Particles to moles Moles and particles · Reading §5.1.3 · 3.5 min

0. **enter** (0.25 min): 9.03×10^{23} iron atoms on the particle side. The arrow points to moles.
1. **predict** (0.25 min): Prediction: multiply or divide? Hands up. Listen for "multiply" (more moles?).
2. **cancel** (0.25 min): The factor flips: 1 mol over 6.022×10^{23} atoms; atoms cancel, mol remains.
3. **result** (0.25 min): 1.50 mol of Fe atoms. Check: 9.03 is 1.5 times 6.022; equal powers of ten.
4. **half a mole** (0.25 min): 3.011×10^{23} molecules is half of 6.022×10^{23}: 0.5000 mol. Misconception: "a count below 6.022×10^{23} cannot be moles".
5. **small and huge** (0.25 min): A sample you can hold is a few moles at most, but an enormous number of particles.
6. **rule** (1 min): Students copy the rule card (reading §5.1.3 Key idea).
7. **check** (0.75 min): Hands up for A, B or C. The half mole stays on the map.
8. **answer** (0.25 min): A is right: $3.011 \times 10^{23} \div 6.022 \times 10^{23} = 0.5000$ mol. B (2.000 mol): that divides 6.022×10^{23} by 3.011×10^{23}, the wrong way around. C (1.813×10^{47} mol): that multiplies by Avogadro's number; particles to moles means dividing.

Reading §5.1.3: particles to moles divides, "the same as multiplying by 1 mol / 6.022×10^{23} particles". Scientific notation: a number from 1 to less than 10 times a power of ten; Lesson 5.3 shows the calculator. 9.03×10^{23} and 6.022×10^{23} have equal powers of ten, so compare 9.03 with 6.022.

18. Formula units Moles and particles · Reading §5.1.3 · 3.25 min

0. **enter** (0.25 min): The NaCl crystal of Unit 4: red + ions and blue − ions, turning slowly. No molecules anywhere.
1. **one formula unit** (0.25 min): One Na^+ and one neighboring Cl^- lit: one formula unit.
2. **predict** (0.25 min): Prediction: 2.71×10^{24} of what? Listen for "molecules".
3. **count** (0.25 min): $4.50 \text{ mol} \times 6.022 \times 10^{23} = 2.71 \times 10^{24}$ formula units (the reading's "On your own").
4. **wrong particle** (0.25 min): "1.51×10^{24} molecules of NaCl": the number is right, the particle is wrong.
5. **rule** (1 min): Students copy the rule card: metals in atoms, molecular compounds in molecules, ionic compounds in formula units.
6. **check** (0.75 min): Hands up for A, B or C. One formula unit is lit again.
7. **answer** (0.25 min): A is right. B (NaCl molecules): sodium chloride is a crystal of ions with no separate molecules. C (sodium atoms and chlorine atoms): in the crystal, sodium and chlorine are ions, not neutral atoms.

Unit 4's formula unit: the group of ions given by the formula of an ionic compound; it states a ratio and is not a molecule. The blueprint's misconception: "2.50 mol of sodium chloride contains 1.51×10^{24} molecules" has the right number ($2.50 \times 6.022 \times 10^{23} = 1.51 \times 10^{24}$) and the wrong particle. Ion sizes from Shannon 1976 (Na^+ 102 pm, Cl^- 181 pm) on the 564 pm cell.

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19. Atoms in a molecule Atoms in a formula · Reading §5.1.4 · 3.5 min

0. **enter** (0.25 min): One H_2O: the subscript 2 is ringed; no subscript after O means one.
1. **predict** (0.25 min): Prediction: one mole of water molecules, how many moles of H atoms? Collect answers. Misconception: "one mole of water holds one mole of hydrogen atoms".
2. **a mole of water** (0.25 min): The subscript multiplies the moles: 1 mol × 2 = 2 mol of H atoms; 1 mol × 1 = 1 mol of O atoms.
3. **carbon dioxide** (0.25 min): CO_2: 2.00 mol × 2 = 4.00 mol of O atoms (the reading's Watch example).
4. **count the atoms** (0.25 min): Then count: $4.00 \text{ mol} \times 6.022 \times 10^{23} = 2.41 \times 10^{24}$ O atoms, twice the 1.20×10^{24} CO_2 molecules.
5. **sodium chloride** (0.25 min): NaCl: 1 mol holds 1 mol of Na^+ ions and 1 mol of Cl^- ions. Read the caption.
6. **rule** (1 min): Students copy the rule card (reading §5.1.4 Key idea).
7. **check** (0.75 min): Hands up for A, B or C. The CO_2 molecule and its table return.
8. **answer** (0.25 min): A is right: two O atoms in every molecule, 2.0 mol × 2. B (2.0 mol): that counts CO_2 molecules; each holds two oxygen atoms. C (1.0 mol): that divides by the subscript; the subscript multiplies.

Reading §5.1.4: in H_2O the subscript 2 means two hydrogen atoms; no subscript after O means one. Molecules in ball and stick: CPK colors, the symbol on every sphere, covalent radii (Cordero 2008), NIST geometries (H_2O 104.5°, CO_2 180°). The NaCl pair uses Shannon radii; the caption under the picture says that atom colors show the element. Also: 0.50 mol of methane, CH_4, holds 2.0 mol of H atoms.

20. Convert: I do, we do, you do Converting · Reading §5.1.3 and §5.1.4 · 6 min

0. **enter** (0.5 min): The steps of every conversion (the reading's form): Given, Factor, Setup, Result, Check.
1. **I do** (0.25 min): I do: 0.250 mol of CO_2 to molecules. Think aloud: the given is in moles, so the factor puts mol below.
2. **cancel** (0.25 min): mol cancels; molecules remain.
3. **result** (0.25 min): 1.51×10^{23} molecules. A quarter of a mole is a quarter of 6.022×10^{23}.
4. **we do** (0.25 min): We do: 1.81×10^{24} water molecules to moles. Students say the factor before the press: 1 mol over 6.022×10^{23} molecules.
5. **cancel** (0.25 min): Students say which unit cancels: molecules.
6. **result** (0.25 min): 3.01 mol. Check: 18.1×10^{23} is about three times 6.022×10^{23}.
7. **you do** (0.25 min): You do: three conversions in the notebook. Read the three rows aloud.
8. **timer 3 min** (3 min): Three minutes of silent work. Circulate.
9. **answer 1** (0.25 min): 0.500 mol × 3 = 1.50 mol of H atoms: the subscript 3 multiplies.
10. **answer 2** (0.25 min): 12.0 mol of H atoms × 6.022×10^{23} = 7.23×10^{24} H atoms.
11. **answer 3** (0.25 min): $2.5 \times 10^{22} \div 6.022 \times 10^{23}$ = 0.042 mol: fewer than 6.022×10^{23} molecules, so less than one mole.

You do: circulate during the timer. Look for "multiply" where dividing is needed (particles to moles), and for results without a named particle. Glucose: 1.00 mol × 12 = 12.0 mol of H atoms, then × 6.022×10^{23} = 7.23×10^{24} H atoms (the reading's "On your own").

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21. The mole in earlier lessons Per mole · Reading §5.1.2 · 2 min

0. **enter** (0.25 min): Lesson 3.2: 496 kJ/mol removes the most loosely held electron from each atom in a mole of gaseous sodium atoms.
1. **predict** (0.25 min): Prediction: per mole means for how many atoms? (6.022×10^{23}.)
2. **one atom** (0.25 min): Divide by 6.022×10^{23}: 8.24×10^{-19} J for one atom.
3. **why per mole** (0.25 min): For one atom the number is tiny, so chemists give such energies per mole.
4. **rule** (1 min): Students copy the rule card (reading §5.1.2 Key idea). No check on this slide.

The reading's Watch example: 496 kJ/mol × 1000 J/kJ × 1 mol / 6.022×10^{23} atoms = 8.24×10^{-19} J per atom. Lesson 3.3 gave heat "per mole of metal" the same way.

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22. Checkpoint routine Checkpoint · Objectives 5.1.2 to 5.1.4 · 1 min

0. **enter** (0.5 min): Explain that the routine repeats for each question.
1. **think** (0.1 min): Think: silent, no pencils.
2. **talk** (0.1 min): Talk: partner A first, then partner B.
3. **write** (0.1 min): Write: the sentence frames and word bank stay on screen.
4. **self-check** (0.1 min): Self-check: tick each criterion you met.
5. **compare** (0.1 min): Compare: read the sample answer and fix one line.

Routine for slides 23 to 25: think 30 s, talk 60 s, write 90 s, self-check 30 s, compare 45 s.

23. Checkpoint 1 Checkpoint · Objective 5.1.2 · 4.5 min

0. **enter** (0.25 min): Read the prompt aloud. Ask: what does the student think a mole is? (a mass).
1. **think 30 s** (0.5 min): Silent thinking. No pencils yet.
2. **talk 60 s** (1 min): Partner A speaks first, then B. Listen for: "a mole weighs one gram" or "6.022 × 10^{23} grams".
3. **write 90 s** (1.5 min): Students write in notebooks with the frames; the word bank stays up.
4. **self-check** (0.5 min): Students tick each criterion they met and fix one line.
5. **sample answer** (0.75 min): Compare with the sample. Reading §5.1.2: a mole is a count, like a dozen; one mole of copper (63.546 g) and one mole of aluminum (26.982 g) hold the same number of atoms.

Routine: think 30 s, talk 60 s, write 90 s, self-check 30 s, compare 45 s. The question stays on screen on every frame.

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24. Checkpoint 2 Checkpoint · Objective 5.1.3 · 4.5 min

0. **enter** (0.25 min): Read the prompt aloud. Two parts: the number, then the particle.
1. **think 30 s** (0.5 min): Silent thinking. No pencils yet.
2. **talk 60 s** (1 min): Partner A speaks first, then B. Listen for: "divide to get particles".
3. **write 90 s** (1.5 min): Students write in notebooks with the frames; the word bank stays up.
4. **self-check** (0.5 min): Students tick each criterion they met and fix one line.
5. **sample answer** (0.75 min): Compare with the sample. Reading §5.1.3: ionic compounds are counted in formula units, one Na^+ ion and one Cl^- ion each.

Routine: think 30 s, talk 60 s, write 90 s, self-check 30 s, compare 45 s. The question stays on screen on every frame.

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25. Checkpoint 3 Checkpoint · Objective 5.1.4 · 4.5 min

0. **enter** (0.25 min): Read the prompt aloud. Ask: what does the small 2 in H_2O tell you?
1. **think 30 s** (0.5 min): Silent thinking. No pencils yet.
2. **talk 60 s** (1 min): Partner A speaks first, then B. Listen for: "the subscript divides".
3. **write 90 s** (1.5 min): Students write in notebooks with the frames; the word bank stays up.
4. **self-check** (0.5 min): Students tick each criterion they met and fix one line.
5. **sample answer** (0.75 min): Compare with the sample. Reading §5.1.4: moles of an element's atoms = moles of molecules × the subscript.

Routine: think 30 s, talk 60 s, write 90 s, self-check 30 s, compare 45 s. The question stays on screen on every frame.

26. Predictions and exit ticket Exit ticket · Objectives 5.1.1 to 5.1.4 · 5.5 min

0. **enter** (0.5 min): Answer 1: the same number of atoms. Hands up if you predicted it. Compare with the counts from slide 4.
1. **answer 2** (0.25 min): Answer 2: formula units. Ask a student who changed their mind to say why.
2. **answer 3** (0.25 min): Answer 3: 2 mol. "1 mol" was the common wrong idea.
3. **exit ticket** (0.5 min): Exit ticket. Claim first: what is a mole?
4. **evidence** (0.25 min): Evidence: the bank's weighing and the step from moles to particles.
5. **write 3 min** (3 min): Reasoning: the subscripts. Three minutes of writing.
6. **self-check** (0.25 min): Self-check with the three criteria.
7. **sample answer** (0.5 min): Compare with the sample. Collect the tickets.

Predictions from slide 4 come back first; students check their letters. Then the CER exit ticket on the driving question, which stays on screen while the sample shows. Collect the exit tickets.

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27. Rate your learning Reflect · All objectives · 1.5 min

0. **enter** (0.5 min): Explain the scale: 1 not yet, 2 almost, 3 I can teach it.
1. **target 1** (0.25 min): Fingers for target 5.1.1. Note who shows 1 and point to the review slides.
2. **target 2** (0.25 min): Fingers for target 5.1.2. Note who shows 1 and point to the review slides.
3. **target 3** (0.25 min): Fingers for target 5.1.3. Note who shows 1 and point to the review slides.
4. **target 4** (0.25 min): Fingers for target 5.1.4. Note who shows 1 and point to the review slides.

Fingers 1 to 3 for each target. Students below 3 revisit the slides named on the right, then the missions: Count by weighing, Moles and particles, Jar puzzles.

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28. Next Next · Missions 5.1 and Lesson 5.2 · 0.5 min

0. **enter** (0.5 min): Point to the three missions. Preview Lesson 5.2: the mass of one mole of any substance, from the periodic table.

Missions for Lesson 5.1: Count by weighing (4 cases), Moles and particles (5 cases), Jar puzzles. Lesson 5.2 finds the mass of one mole of any substance from the periodic table (molar mass).